Logistic Regression Modelling of Credit Card Default Probability and Customer Risk Segmentation: Evidence from Taiwanese Cardholders

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Abstract. Accurate estimation of the probability of default (PD) is central to credit risk management. This study applies logistic regression to the UCI Default of Credit Card Clients dataset (30,000 Taiwanese cardholders, 2005) to predict default in the following month from 20 numerical predictors, using a stratified 80/20 train–test split. The observed default rate was 22.1%, and default rates rose sharply with repayment delay (approximately 13%, 34% and 69% for on-time payment and delays of one and two months). On the test set the model achieved a ROC-AUC of 0.718 and accuracy of 80.4%, but recall of only 21.7% at a 0.50 threshold. Observed default rates increased monotonically across Low, Medium and High risk bands (7.3%, 14.2% and 27.3%), indicating useful rank ordering despite imperfect calibration. The results are illustrative and educational.

Keywords: *credit risk; probability of default; logistic regression; ROC-AUC; model calibration; expected loss*

1. Introduction

The probability of default (PD) is a core input to credit risk management, supporting customer screening, portfolio monitoring and the estimation of expected loss, conventionally expressed as EL = PD × LGD × EAD, where LGD is loss given default and EAD is exposure at default. Credit card portfolios are a demanding setting for PD modelling because exposures are revolving and repayment behaviour changes from month to month. The UCI Default of Credit Card Clients dataset, introduced by Yeh and Lien (2009), is a widely used benchmark for this problem. Logistic regression remains a natural baseline because its coefficients are interpretable and its outputs are probabilities (Hosmer et al., 2013).

This independent educational project, implemented in R, has four objectives: (i) to characterise default behaviour in the customer portfolio; (ii) to estimate PD with a logistic regression model; (iii) to evaluate discrimination and calibration on held-out data; and (iv) to segment customers into risk bands and compute an illustrative expected loss.

2. Methodology

Data. The dataset contains 30,000 credit card customers and 25 variables from a Taiwanese issuer, covering 2005 (Yeh & Lien, 2009). The binary outcome indicates default on the payment in the following month. Data quality checks screened for duplicate records, missing values and invalid target values.

Predictors and split. Twenty numerical predictors were used: credit limit, age, six monthly repayment status variables, six bill amounts and six payment amounts. The customer identifier, the original target column and the demographic variables (sex, education, marital status) were excluded by default. Data were split by stratified sampling into training (80%, n = 24,000) and test (20%, n = 6,000) sets, preserving the default proportion in both; the random seed was 2026.

Model and evaluation. A logistic regression with a binomial distribution and logit link was fitted with R's glm() function. Test-set performance was assessed using ROC-AUC (Fawcett, 2006), a confusion matrix at a 0.50 classification threshold (accuracy, precision, recall, specificity), the Brier score and calibration by decile. Customers were assigned to Low (PD < 5%), Medium (5% to < 15%) or High (≥ 15%) risk bands. Expected loss was computed as PD × LGD × EAD with an assumed LGD of 45% and credit limit as a proxy for EAD; this calculation is illustrative and not a validated estimate of credit losses.

3. Results and Data Interpretation

Of the 30,000 customers, 6,636 defaulted, an observed default rate of approximately 22.1%. Default rates varied by age in a U-shaped pattern: approximately 26.7% for customers aged 25 and below, 19.8% for ages 26–35 (the lowest group) and 26.5% for those above 55. In the fitted model, the strongest predictors, in order of significance, were PAY_0 (most recent repayment status), PAY_AMT1, AGE, BILL_AMT1 and PAY_2. On the test set (threshold 0.50) the model attained ROC-AUC = 0.718, accuracy = 0.804, precision = 0.705, recall = 0.217 and specificity = 0.974.

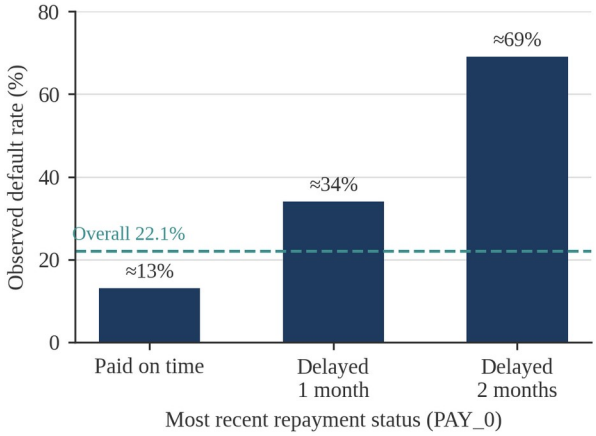


Figure 1. Observed default rate by most recent repayment status (full sample, N = 30,000). The dashed line marks the overall rate.

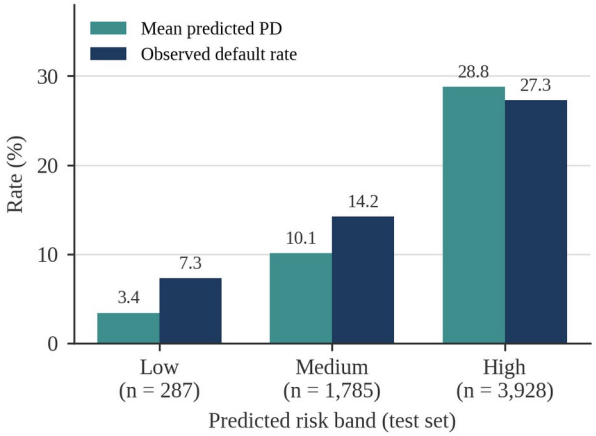


Figure 2. Mean predicted PD and observed default rate by predicted risk band (test set, n = 6,000).

Interpretation of Figure 1. Default rates increase monotonically with the severity of repayment delay, from approximately 13% for customers paying on time to approximately 34% for a one-month delay and 69% for a two-month delay. The two-month-delay rate is more than three times the portfolio rate and roughly five times the on-time rate, consistent with PAY_0 being the leading predictor. These are descriptive associations; category sizes are not reported, so no confidence intervals are given.

Interpretation of Figure 2. Observed default rates rise across the risk bands (7.3%, 14.2%, 27.3%), a spread of about 3.7 times between High and Low, which indicates that the model ranks customers by risk. Calibration is imperfect: mean predicted PD under-states observed default in the Low band (3.4% vs 7.3%) and Medium band (10.1% vs 14.2%) and slightly over-states it in the High band (28.8% vs 27.3%).

4. Discussion

Recent repayment behaviour is the dominant signal of near-term default, which is consistent with the sharp gradient in Figure 1. The model discriminates moderately (AUC = 0.718), and its risk bands order customers correctly, supporting use for ranking rather than for precise probability estimation. Accuracy of 80.4% should be read cautiously: with a 22.1% default rate, a classifier that labelled every customer a non-defaulter would score approximately 77.9% (derived from the base rate). The recall of 21.7% shows that most actual defaulters were missed at the 0.50 threshold, while specificity of 97.4% shows that few non-defaulters were flagged. Because the High-band mean PD (28.8%) lies below 0.50, a lower or cost-sensitive threshold is a plausible remedy, although this was not tested here. The High band also holds 3,928 of 6,000 test customers (about 65%), so the 15% cut-off may be too low to prioritise customers effectively.

Limitations. The 45% LGD and the credit-limit EAD proxy are unvalidated assumptions; no expected loss figure is reported in this paper. A single train–test split was used, without cross-validation, out-of-time testing or stability analysis. The model is a basic logistic regression without feature engineering, and correlated bill and payment amounts call for cautious interpretation of individual coefficients. The 2005 Taiwanese data may not represent current or other populations, and a real credit model would require fairness assessment and regulatory review, particularly as demographic variables were excluded.

5. Conclusion

A transparent logistic regression on 20 behavioural and account variables achieved moderate discrimination (ROC-AUC = 0.718) and produced risk bands whose observed default rates increased from 7.3% to 27.3%, with repayment status as the principal driver. The principal contribution is a reproducible, interpretable PD baseline that also exposes the limits of accuracy as a metric under class imbalance. Future work should tune the threshold using cost-sensitive or precision–recall analysis, add cross-validation, compare regularised logistic and tree-based models, engineer features such as credit utilisation and payment-to-bill ratios, report Gini, KS and Population Stability Index statistics, and perform out-of-time validation.

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